

By defn. of not, $\neg(\neg(x \in A))$ is True. By Double-negation Law of propositions, $x \in A$ is True.

x is an arbitrary element of $\bar{A}$. By Defn. 3 of Section 2.1, $\boxed{\bar{A} \subseteq A}$.

Let $x \in A$. By Defn. of negation, $\neg(x \in A)$ is False. By defn. of not belongs to, $x \notin \bar{A}$ is False. By Defn. 5, $x \in \bar{A}$ is False.

By Defn. of negation, $\neg(x \in \bar{A})$ is True. By Defn. of not belongs to, $x \notin \bar{A}$ is True. By Defn. 5, $x \in \bar{A}$ is True.

x is an arbitrary element of $\bar{A}$. By Defn. 3 of Section 2.1, $\boxed{A \subseteq \bar{A}}$.

$$\therefore \bar{\bar{A}} = A \quad \square$$

e) Prove the identity laws in Table 1 by showing that

$$a) A \cup \phi = A \quad b) A \cap U = A$$

Solution: a) Let x be an arbitrary element of $A \cup \phi$, i.e. $x \in A \cup \phi$.
By Defn. 1, $x \in A \vee x \in \phi$. We know, by defn. of ϕ , $x \notin \phi$. By defn. of not belongs to, $\neg(x \in \phi)$ is True. By Disjunctive syllogism, $x \in A$.
By Defn. 3 of Section 2.1, $\boxed{A \cup \phi \subseteq A}$.

Let, $x \in A$. By Defn. of OR, $x \in A \vee x \in \phi$ is True. By Defn. 1, $x \in A \cup \phi$.
 x is an arbitrary element of A . By Defn. 3 of Section 2.1, $\boxed{A \subseteq A \cup \phi}$.

$$\therefore \boxed{A \cup \phi = A} \quad \square$$

b) ~~Let~~ Let x be an arbitrary element of $A \cap U$, i.e. $x \in A \cap U$.
By Defn. 2, $x \in A \wedge x \in U$. We know, by defn. of AND, $x \in A$ and $x \in U$.
By Defn. 3 of Section 2.1, $\boxed{A \cap U \subseteq A}$.

Let x be an arbitrary element of A , i.e. $x \in A$. Given $A \subseteq U$.
By Defn. 3 of Section 2.1, $x \in U$. By defn. of AND, $x \in A \wedge x \in U$. By Defn. 2, $x \in (A \cap U)$. By Defn. 3 of Section 2.1, $\boxed{A \subseteq A \cap U}$.

$$\therefore \boxed{A \cap U = A} \quad \square$$